

Problem Set 1

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Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

IQ Scores:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

Number of Observations:

```
1 n = length(y)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Descriptive Statistics:

```

1 mean_y <- sum(y)/n
2 sd_y <- sqrt(sum((y - mean_y) ^ 2) / (n - 1))
3 se_y <- sd_y/sqrt(n)

```

Critical Value:

```

1 t_critical <- qt(0.95, df = n - 1)

```

90% Confidence Interval:

```

1 ci_90_lower = mean_y - t_critical * se_y
2 ci_90_upper = mean_y + t_critical * se_y

```

Analysis:

By definition of the confidence interval formula (confidence interval = sample statistic +/- critical value*standard of error), we need to identify our sample statistic, critical value, and margin of error. To calculate the sample statistic, we use the mean of all the observations to identify the trends of all IQ scores on average. This also makes sense because the data is continuous and relatively normally distributed as shown in the visualisation below. Then, for the critical value, a t-critical value would be more suitable than a z-critical value because the sample size of IQ Scores is less than 30 as we observe using the length function (n=25). To find the t-critical value, we can use the built-in qt() function in R, which we identify as 1.711. However, it is important to note that for a 90% confidence interval, the appropriate probability parameter is 0.95 and not 0.90. This follows the idea that we are looking at both tails of the distribution, so when only observing one tail, the area of 0.10 decreases down to 0.05, which translates into a parameter of 0.95. Given that both the population mean and population standard deviation is unknown, we need to subtract one degree from the sample size to receive the degree of freedoms in the second parameter. Then, we need to solve for the standard deviation and square this value to receive the standard error. With both the critical value and standard of error now, we can calculate the margin of error by multiplying the two values. Following the formula for the confidence interval by definition, we can use our information of the sample statistic (sample mean) and add the margin of error to receive the upper bound of the 90% confidence interval. Then, we subtract the margin of error from the sample statistic to receive the lower bound of the 90% confidence interval.

We result in the confidence interval of (93.9599, 102.9201).

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

i. Assumptions (Visualising the Data):

```

1 pdf("plot_1.pdf")
2
3 hist(y,

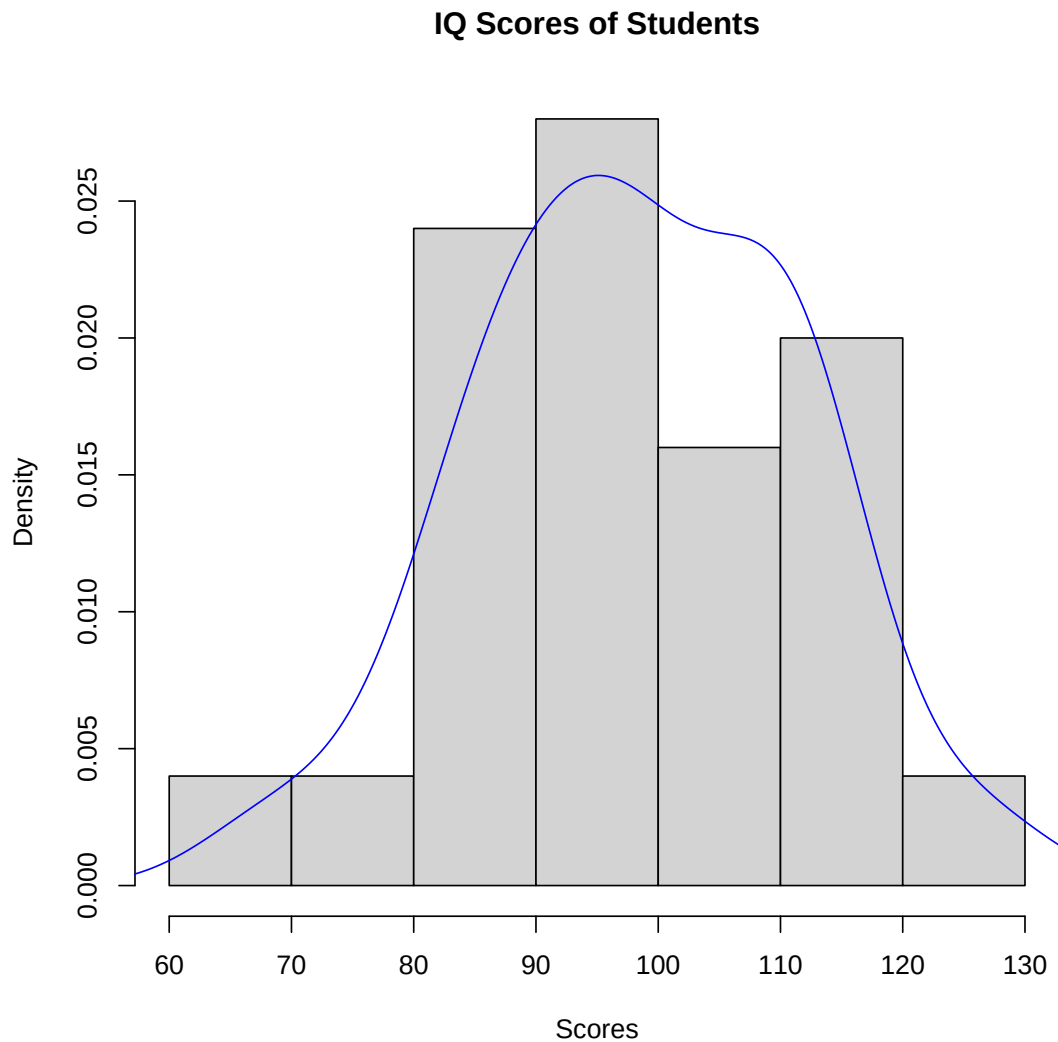
```

```

4     breaks = 8,
5     probability = TRUE,
6     main = "IQ Scores of Students",
7     xlab = "Scores")
8
9 lines(density(y), col = "blue")
10
11 dev.off()

```

Figure 1: IQ Scores of Students



ii. Null and Alternate Hypotheses:

Question: Is the average IQ score in the counselor's school higher than the average IQ score of 100 among all schools in the country?

$$H_0 : \mu = 100 \quad \text{(the average IQ score is 100)}$$

$$H_A : \mu > 100 \quad \text{(the average IQ score is greater than 100)}$$

iii. *Test Statistic:*

```
1 mu_0 = 100
1 t_score <- (mean_y - mu_0) / se_y
1 t_critical <- qt(0.95, df = n - 1)
```

iv. *P-value:*

```
1 p_value <- pt(t_score, df = n-1, lower.tail = FALSE)
```

Check Using the Built-in t.test() Function:

```
1 t.test(y, mu = 100, alternative = "greater", conf.level = 0.95)
```

v. *Conclusion:*

Analysis:

To figure out whether the average student IQ in the counselor's school is higher than the average IQ score among all schools in the country, we proceed with a one-tailed significance test. Following the 5 steps of hypothesis testing, we first identify whether the sample meets the assumption criterias. We know that IQ is a continous variable, measured on an interval scale (contains a true zero), and that the sample size is 25. Information given to us include that the sample is random. The visualisation of the sample distribution below also illustrates that the data relatively follows a normal distribution from its bell-shaped curve. This is important to emphasize because the sample size is less than 30, so we cannot use the central limit theorem and a large amont of sampling instances to determine normality. Lastly, we can assume that the IQ scores in the sample are independent of each other because they are drawn from a school population, where most students are not from the same family such that genetic similarities would cause dependency across observations.

After verifying that all the assumptions are met, we formulate the null and alternate hypothesis. The null hypothesis states that there is no difference between the IQ score within the counselor's school and among all schools in the country, such that both on average is 100. In contrast, the alternate hypothesis states that the average IQ score is greater than that of students in all schools in the country.

There are two ways we can come to the right conclusion. The first method would be to calculate the t-critical value and t-statistic. Again, the t-test is used because the sample size is less than 30 and both the population mean and standard deviation are unknown. For the alpha threshold of 0.05, we enter 0.95 as the probability in the first parameter of the built-in qt() functino to calculate the t-critical value since we are only observing a one-tail test. If the t-statistic is significantly greater than the t-critical value, then

the difference is highly unlikely to be a random occurrence, allowing us to reject the null hypothesis. However, given that the t-statistic value is -0.59574 and the t-critical value is 1.71088, the t-statistic value is significantly lower than the t-critical value. This indicates that any differences within the sample could occur from random chance, so we fail to reject the null hypothesis. The p-value corroborates this conclusion, but simply provides a second strategy to approach the significance test. The p-value we observe is 0.7215, which is significantly greater than the alpha threshold of 0.05. Hence, this indicates that in a world where the null is true, it is unlikely that we observe a value more extreme than the observed value. With that in mind, we would fail to reject the null hypothesis.

We fail to reject the null hypothesis and can conclude that there is not sufficient evidence to say that the average student IQ in the counselor's school is higher than the average IQ score (100) among all the schools in the country.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

Import Data:

```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

Explore Data:

```
1 head(expenditure)
2 str(expenditure)
3 summary(expenditure)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

Define Variables:

```
1 vars <- c("Y", "X1", "X2", "X3")
```

Variable Labels:

```
1 labels <- c(
2   "Expenditure",
3   "Income",
4   "Financially Insecure",
5   "Urban"
6 )
```

Set Up Plot:

```
1 pdf("plot_2.pdf", width = 21, height = 12)
2
3 par(mfrow = c(2, 3), cex = 1.8)
```

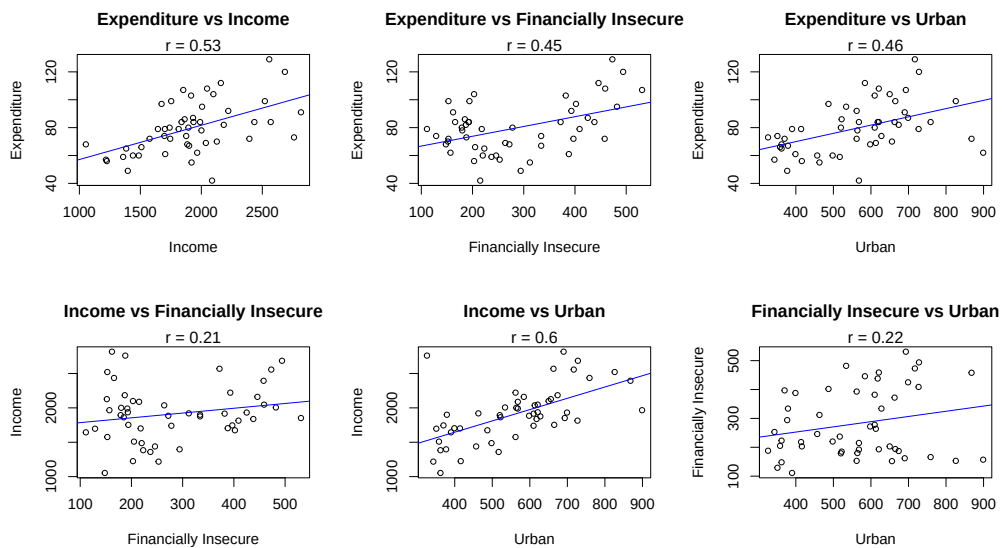
Pairwise Scatterplots and Correlations:

```

1 for (i in 1:(length(vars) - 1)) {
2   for (j in (i + 1):length(vars)) {
3
4     plot(
5       expenditure[[vars[j]]],
6       expenditure[[vars[i]]],
7       xlab = labels[j],
8       ylab = labels[i],
9       main = paste(labels[i], " vs", labels[j]),
10      cex = 0.8
11    )
12
13    abline(
14      lm(expenditure[[vars[i]]] ~ expenditure[[vars[j]]]),
15      col = "blue",
16    )
17
18    r <- cor(
19      expenditure[[vars[i]]],
20      expenditure[[vars[j]]]
21    )
22
23    mtext(
24      paste("r =", round(r, 2)),
25      cex = 2
26    )
27  }
28 }
29
30 dev.off()

```

Figure 2: Pairwise Relationships and Correlations



Analysis:

To plot the pairwise scatterplots between each respective variable, I chose to use a nested for loop inside another for loop because the framework for each graph essentially stays the same, but the labels, data points, line of best fit, and correlations change amongst them. The outer loop selects each variable in turn, while the inner loop compares it with every variable that follows it. This produces all six unique pairwise comparisons without plotting a variable against itself or repeating the same relationship twice. A scatterplot deems the most representative graph in this case because it most accurately represents the correlation between two continuous variables, displaying the magnitude and direction with the line of best fit. Scatterplots also efficiently visualise any outliers that may be skewing the data.

The three scatterplots in the first row, comparing the explanatory variables (income, financially insecure, and urbanisation) each respectively against the response variable (expenditure), all have a Pearson correlation coefficient r that range from 0.45 to 0.53. This indicates that expenditure has a positive moderate linear relationship with income, "financially insecure" residents, and individuals who reside in urban areas. In other words, in general, higher per capita income, levels of financial insecurity, and urban population are each associated with higher per capita housing expenditure.

Proceeding to the next three scatterplots in the second row, where we observe the correlations between the explanatory variables, the correlation coefficients of financially insecure with income and urbanisation both have weak positive linear correlations as seen from the r coefficient of 0.21 and 0.22 respectively. We also see this since in both graphs, the observations are widely dispersed around the fitted line. Lastly, income and individuals who reside in urban areas have the strongest positive linear relationship as we observe from the correlation coefficient of 0.6.

Several potential outliers can be identified within each panel, particularly unusually low or high expenditure observations that deviate from the fitted lines in scatterplots with strong or moderate positive linear correlations. For example, the expenditure plots include an unusually low expenditure near 42 and several observations above 120. In contrast, the weaker relationships, income vs. financially insecure and urban vs. financially insecure, show broader dispersion around the fitted line rather than clearly isolated outliers, suggesting that their weaker correlations reflect greater overall variability rather than a few unusual observations.

Overall, housing expenditure is moderately strong and positively associated with income, financial insecurity, and urbanisation, while income and urbanisation show the strongest positive linear relationship and financial insecurity has only weak positive linear relationships with income and urbanisation.

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

Define Regions:

```

1 regions <- factor(
2   expenditure$Region,
3   levels = c(1, 2, 3, 4),
4   labels = c("North East", "North Central", "South", "West")
5 )

```

Compare Regional Means and Medians:

```

1 for (i in levels(regions)) {
2   region_mean <- round(mean(expenditure$Y[regions == i]), 2)
3   region_median <- round(median(expenditure$Y[regions == i]), 2)
4
5   print(paste(i, "mean expenditure =", region_mean))
6   print(paste(i, "median expenditure =", region_median))
7   cat("\n")
8 }

```

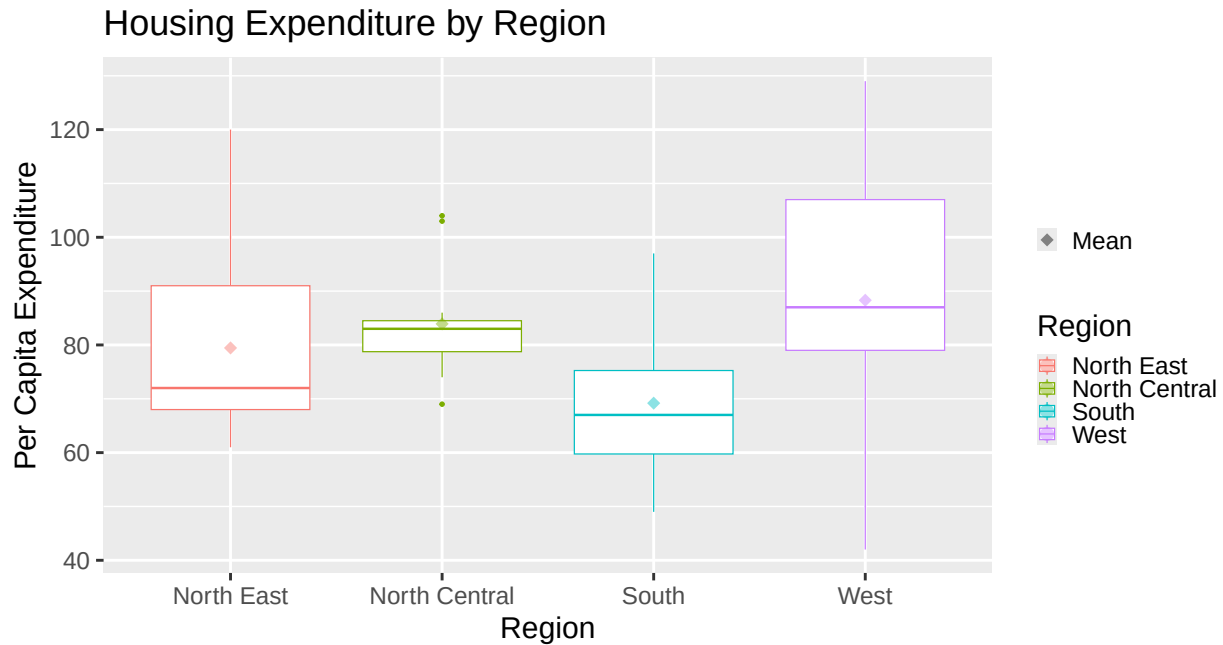
Plot Housing Expenditure by Region:

```

1 plot_3 <- ggplot(expenditure, aes(x = regions, y = Y, color = factor(
2   regions))) +
3   geom_boxplot() +
4   stat_summary(
5     aes(shape = "Mean"),
6     fun = mean,
7     geom = "point",
8     size = 5,
9     alpha = 0.45
10  ) +
11  labs(
12    title = "Housing Expenditure by Region",
13    x = "Region",
14    y = "Per Capita Expenditure",
15    color = "Region",
16    shape = ""
17  ) +
18  scale_shape_manual(
19    values = c("Mean" = 18)
20  ) +
21  theme_gray(base_size = 26)
22 ggsave("plot_3.pdf", plot = plot_3, width = 15, height = 8)

```

Figure 3: Housing Expenditure by Region



Analysis:

Because the raw dataset provides the regions as numerical values, I began by labelling each respective number with its associated region name with a for loop.

Then, I proceeded with a boxplot because it best compares data distributions across multiple groups, while spotting outliers. The boxplot also displays the median, which is a better measure of central tendency in comparison to the mean if there are outliers present. However, observing the boxplots and calculations of per capita expenditure for each region, there seems to be a negligible difference between the mean and median, since there are not many outliers observed other than the three potential observations in the North Central region. This further validates the decision for using a boxplot to assist in determining which region has the highest per capita expenditure on housing assistance.

Comparing the means and medians, the West region surpasses all the other three regions in per capita expenditure. Even the first quartile of expenditure in the West is higher than the medians of the North East and South and close to that of the North Central, further portraying that there is generally a higher level of per capita expenditure in the West.

Numerically, we see this result when comparing the North East mean (\$79.44) and median (\$72) against the North Central mean (\$83.92) and median (\$83) as well as against the South mean (\$69.19) and median (\$67). The West has a central tendency in expenditure much higher in comparison to all three with a mean of \$88.31 and median of \$87.

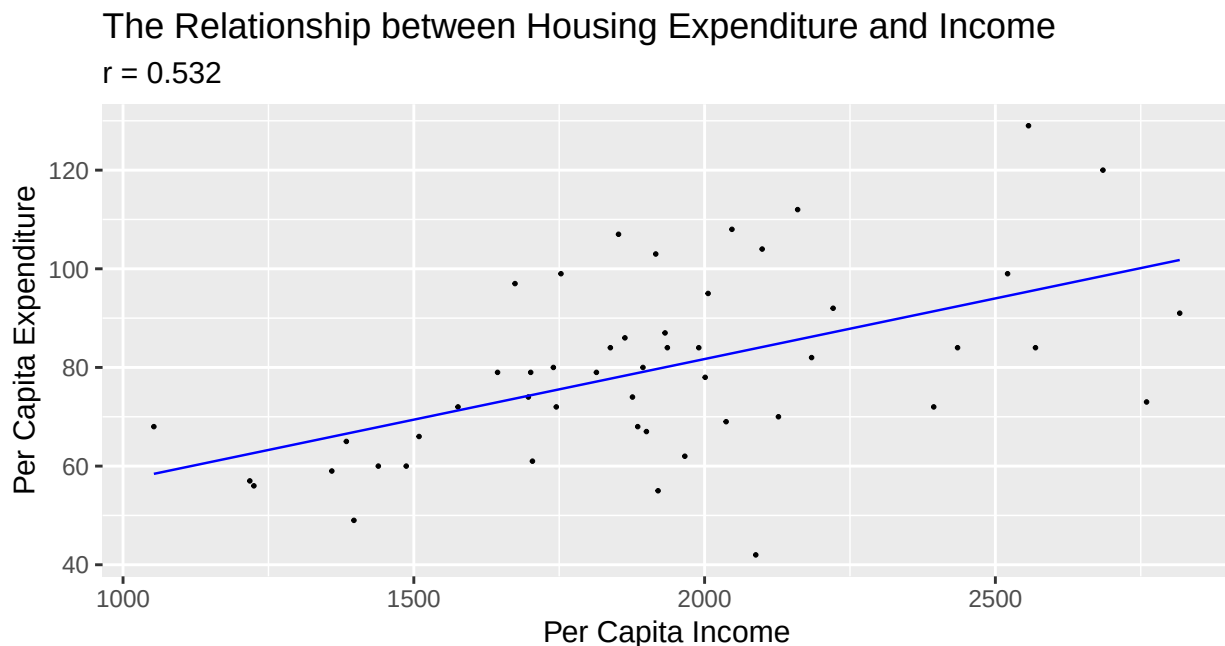
On average, the West region has the highest per capita expenditure on housing assistance, with a mean of \$88.31 and median of \$87.

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

Housing Expenditure and Income:

```
1 plot_4 <- ggplot(expenditure, aes(x = X1, y = Y)) +  
2   geom_point() +  
3   geom_smooth(  
4     method = "lm",  
5     color = "blue",  
6     se = FALSE  
7   ) +  
8   labs(  
9     title = "The Relationship between Housing Expenditure and Income",  
10    subtitle = paste("r =", round(cor(expenditure$X1, expenditure$Y), 3))  
11  ,  
12    x = "Per Capita Income",  
13    y = "Per Capita Expenditure"  
14  ) +  
15  theme_gray(base_size = 26)  
16 ggsave("plot_4.pdf", plot = plot_4, width = 15, height = 8)
```

Figure 4: Housing Expenditure versus Income



Analysis:

To analyse the relationship between per capita housing expenditure and per capita income, a scatterplot would be the best representation since the two variables are continuous. The line of best fit would allow for us to observe the magnitude and direction of the association between the two variables. We observe a moderate positive linear relationship between per capital income and per capita expenditure, also represented by the correlation coefficient of 0.532. There are several outliers that stand out, since they noticeably deviate from the overall linear trend. For example, there is one observation around an income of 2,100 with expenditure near 42, and another around an income of 2,750 with expenditure near 73. There are also a few unusually high expenditure observations around incomes of 2,550 to 2,700, with expenditures above 120. However, because most points cluster near the line of best fit and there is a moderate positive relationship, these outliers would be unlikely to affect the overall correlation.

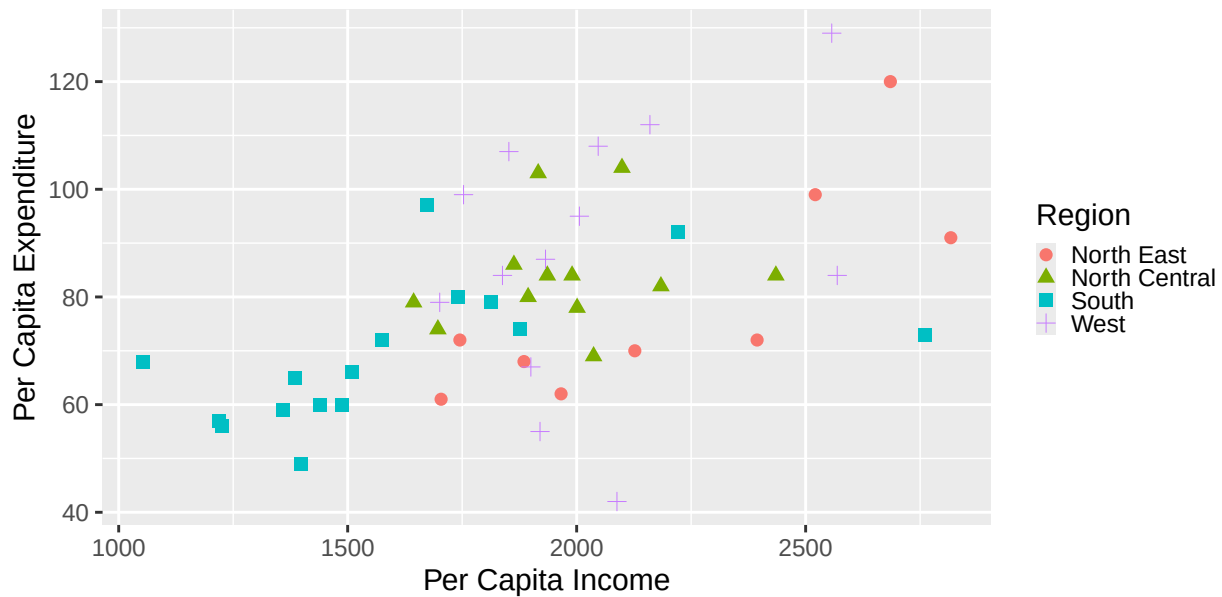
Overall, per capita housing expenditure and per capita income have a moderately positive linear relationship. In general, higher per capita income is associated with higher housing expenditure.

Housing Expenditure and Income by Region:

```
1 plot_5 <- ggplot(  
2   expenditure ,  
3   aes(  
4     x = X1,  
5     y = Y,  
6     color = factor(regions) ,  
7     shape = factor(regions)  
8   )  
9 ) +  
10 geom_point(size = 5) +  
11 labs(  
12   title = "The Relationship between Housing Expenditure and Income by  
13   Region" ,  
14   x = "Per Capita Income" ,  
15   y = "Per Capita Expenditure" ,  
16   color = "Region" ,  
17   shape = "Region"  
18 ) +  
19 theme_gray(base_size = 26)  
20 ggsave("plot_5.pdf" , plot = plot_5, width = 15, height = 8)
```

Figure 5: Housing Expenditure versus Income by Region

The Relationship between Housing Expenditure and Income by Region



Correlations by Region:

```
1 for (i in levels(regions)) {
2   region_correlation <- round(cor(expenditure$X1[regions == i],
3     expenditure$Y[regions == i]), 3)
4   print(paste(i, "correlation =", region_correlation))
5   cat("\n")
}
```

Analysis:

When the observations are separated based on region, we observe that some regions have much stronger positive linear correlation between the two variables than other regions. This insinuates that the aggregated data masks underlying relationships that can only be observed within certain groups based on region. For example, in the South, individuals tend to have both lower income and lower expenditure. In contrast, individuals in the North East, North Central, and West exhibit higher levels of both income and expenditure. When the observations are separated by region, we observe less of a linear association between the data points in the North Central and West. Using a for loop, we can also identify the Pearson correlation coefficients for each region. We observe the respective correlation coefficients of $r=0.184$ for North Central and $r=0.305$ for West, which are much lower in comparison to the other two regions. In contrast, the data points in the South have a moderate positive linear correlation at $r=0.556$ and the data points in the North East have a strong positive linear correlation at $r=0.802$. In general, increases in per capita income are associated with increases

in per capita housing expenditure, although this relationship is much stronger in the North East and South than in the North Central and West.

Overall, when grouped by region, the overall relationship between income and expenditure differs across the regions, with the South concentrated at lower income and expenditure levels and substantially stronger correlations in the North East and South than in the North Central and West.